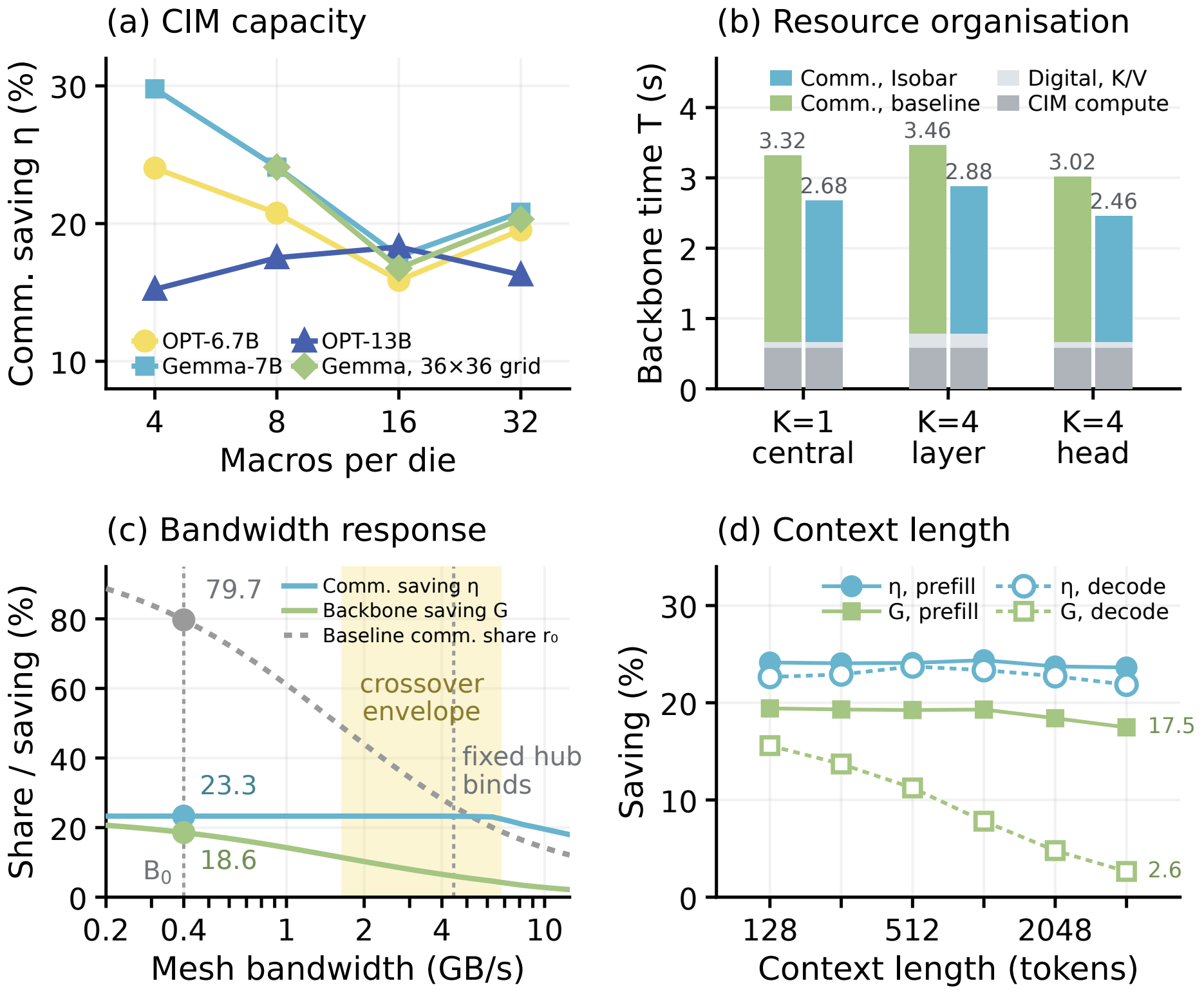
(a) CIM capacity
Comm. saving η (%)
30
20
10
OPT-6.7B
OPT-13B
Gemma-7B
Gemma, 36×36 grid
4
8
16
32
Macros per die
(b) Resource organisation
Backbone time T (s)
4
3
2
1
0
Comm., Isobar
Comm., baseline
Digital, K/V
CIM compute
3.32
2.68
3.46
2.88
3.02
2.46
K=1
central
K=4
layer
K=4
head
(c) Bandwidth response
Share / saving (%)
80
60
40
20
0
79.7
Comm. saving η
Backbone saving G
Baseline comm. share r₀
crossover
envelope
fixed hub
binds
23.3
18.6
B₀
0.2
0.4
1
2
4
10
Mesh bandwidth (GB/s)
(d) Context length
Saving (%)
30
20
10
0
η, prefill
η, decode
G, prefill
G, decode
17.5
2.6
128
512
2048
Context length (tokens)